

WUTAI SHAN, China

WMO: 535880

Lat: 38.9417N

Lon: 113.5133E

Elev: 7251

StdP: 11.23

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-5.6	-30.3	1.3	2.8	-25.7	1.7	4.2	52.9	-9.7	44.8	-8.6	23.4	270	0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	70.1	57.4	68.1	56.7	66.1	55.8	63.0	66.1	61.2	64.4	59.6	62.9	10.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.1	109.5	64.4	60.1	102.2	62.9	58.4	95.8	61.5	32.8	66.1	31.2	64.6	29.8	63.0	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	38.3	32.3	26.9	DB	-16.1	74.8	6.4	3.2	-20.6	77.1	-24.3	79.0	-27.9	80.8	-32.5	83.1
(5)				WB	-16.6	65.6	6.2	2.0	-21.1	67.1	-24.8	68.2	-28.3	69.4	-32.8	70.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	10.6	15.2	25.4	38.2	48.3	55.3	59.5	56.9	49.4	38.4	24.4	13.1	(6)
(7)		DBStd	18.82	8.94	8.98	9.59	9.24	6.90	5.13	4.22	4.59	5.58	7.41	8.83	8.61	(7)
(8)		HDD50	5813	1223	975	763	364	114	15	0	6	76	363	769	1145	(8)
(9)		HDD65	10473	1688	1395	1228	804	517	291	176	251	469	825	1219	1610	(9)
(10)		CDD50	822	0	0	0	11	61	175	294	221	57	3	0	0	(10)
(11)		CDD65	7	0	0	0	0	0	1	5	1	0	0	0	0	(11)
(12)		CDH74	11	0	0	0	0	0	3	8	0	0	0	0	0	(12)
(13)		CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	12.2	14.4	14.1	14.1	13.7	12.2	10.2	8.8	8.7	9.2	11.8	14.4	15.3	(14)
(15)	Precipitation	PrecAvg	20.6	0.2	0.4	0.6	1.0	1.6	3.0	5.1	4.3	2.6	1.1	0.6	0.2	(15)
(16)		PrecMax	26.7	0.7	1.1	2.2	2.4	2.8	7.1	10.0	9.6	4.6	4.0	2.0	0.9	(16)
(17)		PrecMin	13.8	0.0	0.0	0.0	0.3	0.4	1.3	2.5	1.5	0.7	0.2	0.0	0.0	(17)
(18)		PrecStd	3.2	0.2	0.2	0.6	0.5	0.6	1.3	1.7	1.7	1.1	0.8	0.6	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	39.6	53.1	65.1	69.2	72.3	74.1	70.8	66.3	58.8	47.3	35.7	(19)
(20)		0.4%	MCWB	24.2	28.8	37.0	44.5	48.6	53.0	61.0	59.9	52.0	45.5	34.0	25.4	(20)
(21)		2%	DB	29.3	34.3	47.2	59.4	65.1	68.5	70.8	68.2	62.7	54.7	42.6	30.3	(21)
(22)		2%	MCWB	20.9	25.6	33.3	41.9	47.3	53.0	59.4	58.9	51.4	43.6	31.7	22.2	(22)
(23)		5%	DB	25.1	30.5	42.7	55.4	62.1	66.1	68.7	66.2	60.0	51.6	39.2	27.0	(23)
(24)		5%	MCWB	18.6	23.7	31.2	39.7	45.6	52.7	58.6	58.3	50.1	41.5	30.6	20.6	(24)
(25)		10%	DB	21.9	27.4	38.7	51.3	59.1	63.7	66.6	64.1	57.5	48.5	36.0	24.1	(25)
(26)		10%	MCWB	16.3	21.4	29.0	38.0	44.3	51.6	58.1	57.5	49.3	39.9	29.1	18.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	30.2	38.7	49.1	53.9	59.8	66.0	65.1	58.1	49.4	37.2	26.5	(27)
(28)		0.4%	MCDWB	33.4	34.9	47.5	54.9	60.8	65.3	70.0	68.2	61.3	55.0	41.1	31.8	(28)
(29)		2%	WB	21.5	26.9	34.7	45.2	51.0	57.6	63.8	62.8	55.3	46.1	34.2	23.4	(29)
(30)		2%	MCDWB	27.8	32.4	45.1	55.1	59.2	63.2	67.0	65.3	58.1	51.7	39.3	29.5	(30)
(31)		5%	WB	19.0	24.2	31.9	42.2	49.0	56.0	62.0	61.0	53.0	43.7	31.7	20.9	(31)
(32)		5%	MCDWB	24.7	30.1	41.6	52.5	57.4	61.8	65.4	63.6	56.9	49.1	37.8	26.3	(32)
(33)		10%	WB	16.7	21.6	29.6	39.7	46.9	54.3	60.4	59.4	51.1	41.4	29.4	18.8	(33)
(34)		10%	MCDWB	21.6	27.2	38.0	48.9	55.3	60.1	63.9	62.1	55.4	46.6	35.5	23.8	(34)
(35)	Mean Daily Temperature Range		MDBR	11.7	12.2	13.7	14.8	14.5	12.7	10.6	10.4	10.8	11.8	11.7	11.3	(35)
(36)		5% DB	MCDBR	13.4	14.3	16.7	18.4	17.2	15.9	13.4	12.6	13.0	14.3	13.5	12.7	(36)
(37)		5% DB	MCWBR	9.5	10.1	10.0	9.7	9.7	9.2	9.3	8.9	8.8	8.5	8.8	9.0	(37)
(38)		5% WB	MCDBR	12.8	13.4	16.2	16.9	15.2	13.4	11.3	10.6	11.0	12.4	12.3	12.0	(38)
(39)		5% WB	MCWBR	9.7	10.0	10.5	10.4	9.8	9.3	9.1	8.3	8.9	8.5	8.8	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.342	0.359	0.400	0.431	0.516	0.508	0.496	0.439	0.387	0.306	0.281	(40)	
(41)		taud	2.313	2.150	2.143	2.089	2.038	1.855	1.925	1.933	2.035	2.115	2.318	2.364	(41)	
(42)		Ebn at Noon	281	275	283	278	270	247	247	247	253	254	270	275	(42)	
(43)		Edh at Noon	29	39	44	50	54	64	60	58	49	40	29	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	853	1079	1492	1749	1937	1723	1590	1508	1317	1109	871	763	(44)	
(45)		RadStd	70	84	123	116	156	141	94	124	104	96	112	67	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	-3.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	-1.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page